Air Quality Index (AQI) Prediction System

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10Pearls

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1 Introduction

The Air Quality Index (AQI) is a standardized indicator that quantifies the concentration of major air pollutants such as $PM_{2.5}$, PM_{10}, O_3, NO_2, SO_2, and CO. This project aims to build a fully automated and production-ready AQI prediction system capable of:

- Collecting real-time data from World Air Quality Index (WAQI) API,
- Cleaning and processing the data through a robust 10-step pipeline,
- Engineering multiple predictive features,
- Training and comparing multiple ML models, and
- Deploying predictions through a web-based dashboard.

2 Objectives

The main objectives of the system are:

1. Automate hourly data collection from the WAQI API.
2. Develop a reliable data preprocessing and cleaning pipeline.
3. Perform advanced feature engineering for time-series forecasting.
4. Train and evaluate multiple machine learning models for AQI prediction.
5. Deploy the best model for real-time inference using a Streamlit dashboard.
6. Integrate MLOps practices with Hopsworks Feature Store and GitHub Actions.

3 System Architecture

The overall architecture integrates data collection, model training, feature storage, and deployment in a cohesive MLOps pipeline.

3.1 Data Flow

WAQI API $\longrightarrow$ Data Fetcher $\longrightarrow$ Data Cleaning $\longrightarrow$ Feature Engineering $\longrightarrow$ Hopsworks Feature Store $\longrightarrow$

3.2 Components

Table 1: System Components and Descriptions

Component	Description
Data Fetcher	Collects hourly air quality and weather data using WAQI API.
Data Cleaning	Applies a 10-step cleaning pipeline for high-quality input data.
Feature Engineering	Generates multiple lag, rolling, and domain-specific features.
Model Training	Compares XGBoost, Random Forest, LSTM, and CNN models.
Feature Store	Manages features and ensures training-serving consistency.
Streamlit Dashboard	Provides a real-time, interactive visualization of predictions.
GitHub Actions	Automates hourly data updates and predictions.

4 Methodology

4.1 Data Collection

Data was fetched every hour via the World Air Quality Index (WAQI) API. Collected pollutants and weather features include:

- $PM_{2.5}$, PM_{10}, O_3, NO_2, SO_2, CO
- Temperature, Humidity, Pressure, Wind Speed

4.2 Data Cleaning Pipeline

A comprehensive 10-step cleaning pipeline was implemented:

1. Duplicate removal
2. AQI range validation (0–500)
3. Pollutant-level verification
4. Outlier detection (IQR + Z-score)
5. Domain-based value capping
6. Temporal consistency checks
7. Feature correlation analysis
8. Low variance feature removal
9. Rolling median smoothing
10. Final validation and integrity check

Outcome: 85–92% data retention with $>$95% quality assurance.

5 Machine Learning Models

Four models were trained and evaluated for comparative performance:

Table 2: Model Performance Comparison

Rank	Model	RMSE	MAE	R^2	Performance
1	XGBoost	9.44	5.55	0.947	Excellent
2	Random Forest	9.59	5.70	0.945	Excellent
3	LSTM	16.34	10.90	0.842	Good
4	CNN 1D	45.42	38.53	-0.220	Poor

XGBoost achieved 94.7% variance explanation and was chosen for deployment.

6 Model Details

6.1 XGBoost Configuration

```
XGBRegressor(
```

```
    n_estimators=100,
    learning_rate=0.1,
    max_depth=6,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42
)
```

6.2 Cross-Validation

- **Method:** TimeSeriesSplit (5-fold)
- **Metrics:** R^2, RMSE, MAE
- **Result:** Mean $R^2 = 0.931 \pm 0.023$, Mean RMSE $= 7.51 \pm 2.92$

7 MLOps & Deployment

7.1 Hopsworks Feature Store

- Centralized repository for engineered features
- Version-controlled and time-travel capable
- Ensures feature consistency between training and production

7.2 GitHub Actions Automation

Frequency: Every hour

Tasks Automated:

1. Environment setup
2. Data collection and cleaning
3. Feature generation and upload
4. Model inference
5. JSON update for dashboard visualization

7.3 Streamlit Dashboard

- Displays current AQI with color-coded health categories
- Provides 3-day forecast with trend visualization
- Showcases model analytics and feature importance

8 Results and Discussion

The XGBoost model demonstrated superior performance with:

- **RMSE:** 9.44
- **MAE:** 5.55

- **R^2:** 0.947

This means the model can predict AQI within ± 9 points of actual values on average, representing strong predictive capability. The feature analysis confirmed the dominance of $PM_{2.5}$-related lag and rolling mean features, consistent with domain understanding that $PM_{2.5}$ is a primary driver of air quality fluctuations.

9 Conclusion

This project successfully demonstrates a production-grade AQI prediction system integrating machine learning, MLOps, and data automation.

By achieving 94.7% prediction accuracy, the system provides reliable air quality forecasts and forms a strong foundation for scalable environmental monitoring platforms.

With further improvements and broader data integration, this approach can support smart city air quality management and public health decision-making.

Streamlit Deployed Link: https://airqualityindexprediction.streamlit.app/

10 References

1. EPA Air Quality Index Guide: https://www.airnow.gov/aqi/aqi-basics/
2. WAQI API Documentation: https://aqicn.org/api/
3. Hopsworks Documentation: https://docs.hopsworks.ai/
4. Scikit-learn Documentation - Time Series Cross-Validation.
5. YouTube videos provided in Discord server.
6. ChatGPT for learning new techniques and to cater low R^2 scores.